21-242 Recitation 12

Span, Linear Independence, Bases — Q3. Independence iff only the trivial combination is

Course-level solution, with a later-course alternative when useful

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Prompt

Prove that $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n \subseteq V$ is linearly independent if and only if $\sum_{i=1}^{n} a_i \vec{v}_i = \vec{0}$ implies $a_i = 0$ for all i.

Course-level solution

Write $\mathcal{L} = (\vec{v}_1, \ldots, \vec{v}_n)$. We use the sequential definition from Q2.
($\Rightarrow$). Suppose $\mathcal{L}$ is linearly independent and $\sum_{i=1}^{n} a_i \vec{v}_i = \vec{0}$. If some coefficient is nonzero, let j be the *largest* index with $a_j \neq 0$. Then

$$\vec{v}_j = \sum_{i=1}^{j-1} \Big(-\frac{a_i}{a_j}\Big) \vec{v}_i \in \operatorname{Span}(\vec{v}_1, \ldots, \vec{v}_{j-1})$$

(if $j = 1$ this says $\vec{v}_1 = \vec{0}$). This contradicts sequential independence. Hence every $a_i = 0$.
($\Leftarrow$). Suppose the only linear combination equal to $\vec{0}$ is the trivial one. If the list failed sequential independence, some $\vec{v}_j$ would lie in $\operatorname{Span}(\vec{v}_1, \ldots, \vec{v}_{j-1})$, say $\vec{v}_j = \sum_{i<j} c_i \vec{v}_i$. Then

$$\sum_{i<j} c_i \vec{v}_i + (-1)\vec{v}_j + \sum_{i>j} 0 \cdot \vec{v}_i = \vec{0}$$

is a nontrivial combination (the coefficient of $\vec{v}_j$ is -1), a contradiction. Thus $\mathcal{L}$ is linearly independent.

Later-course solution (using ideas from the rest of the handout)

After bases and dimension: a list of length n in an n-dimensional space is independent iff it is spanning iff it is a basis, and the trivial-combination criterion is the statement that the map $\mathbb{F}^n \to V$, $e_i \mapsto \vec{v}_i$, is injective. That is a cleaner packaging, but it uses the linear-map language from the next section.